BIOLOGY / 18.1 · STUDENT FIELD WORKBOOK

Variation Lab

Continuous and discontinuous variation

Eight investigations for measuring, recording, plotting, and thinking carefully



Cover picture: the Variation Lab Live website header.

A blank, print-ready workbook for offline classroom use. No name, student ID, or class code is needed. Participation is optional.

BEFORE YOU BEGIN

One class. Many patterns.

OVERALL LEARNING OBJECTIVE

By the end, I can measure or classify variation safely, record anonymous evidence, choose a suitable graph, plot group frequencies, and explain what the sample can and cannot show.

Keep participation safe and optional

You may pass on any body measurement or category without giving a reason. A teacher-provided practice dataset is a complete alternative. Do not write names, student IDs, diagnoses, or other identifying details. Keep individual body measurements private; show only pooled class totals.

INVESTIGATION 6 IS A SIMULATION ONLY

No real blood, needles, lancets, or bodily fluids.

Never ask anyone to disclose a real blood group.

Choose the graph that matches

Continuous measurements sit on a numerical scale: use a histogram, choose equal-width intervals, and make the bars touch. **Categories** are separate groups: use a bar chart with gaps between bars. Label both axes and start the frequency scale at zero.

For leaves and seeds, use only teacher-approved materials, check school allergy rules, do not eat seeds, and wash hands after handling plant material.

Inside this workbook

No.	Investigation	Workbook pages
01	Body mass · kg	01–02
02	Height · cm	03–04
03	Hand span · cm	05–06
04	Tongue rolling · Can/Cannot	07–08
05	Eye colour · Brown/Blue/Green/Hazel	09–10
06	Simulated blood group · A/B/AB/O	11–12
07	Leaf length · cm	13–14
08	Seed mass · g	15–16

The booklet is a blank template. It is not connected to the live dashboard and contains no class submissions.

INVESTIGATION 01 / METHOD + RECORD

Body mass

LEARNING OBJECTIVE

Measure body mass in kilograms (kg) and use anonymous, voluntary class data to describe its distribution with a histogram.



Image: Variation Lab Live website

Method

1. Participation is optional; use a teacher dataset instead if you prefer.
2. Set an approved scale on a level, non-slip floor in a private area.
3. If you choose, measure only yourself and step on/off as directed.
4. Record anonymous kilograms (20–300 kg); keep the value private. Never rank, name, or change a result.

Risk assessment Hazard → safe action

Sensitive measure	Make it optional and private; discuss pooled totals only.
Slip or trip	Use a level, non-slip surface; clear the area and step carefully.

Record your data Up to 40 anonymous readings · 20–300 kg. Row numbers are not people; report pooled intervals only.

No.	Mass (kg)	No.	Mass (kg)
1		21	
2		22	
3		23	
4		24	
5		25	
6		26	
7		27	
8		28	
9		29	
10		30	
11		31	
12		32	
13		33	
14		34	
15		35	
16		36	
17		37	
18		38	
19		39	
20		40	

INVESTIGATION 01 / GRAPH + INTERPRET

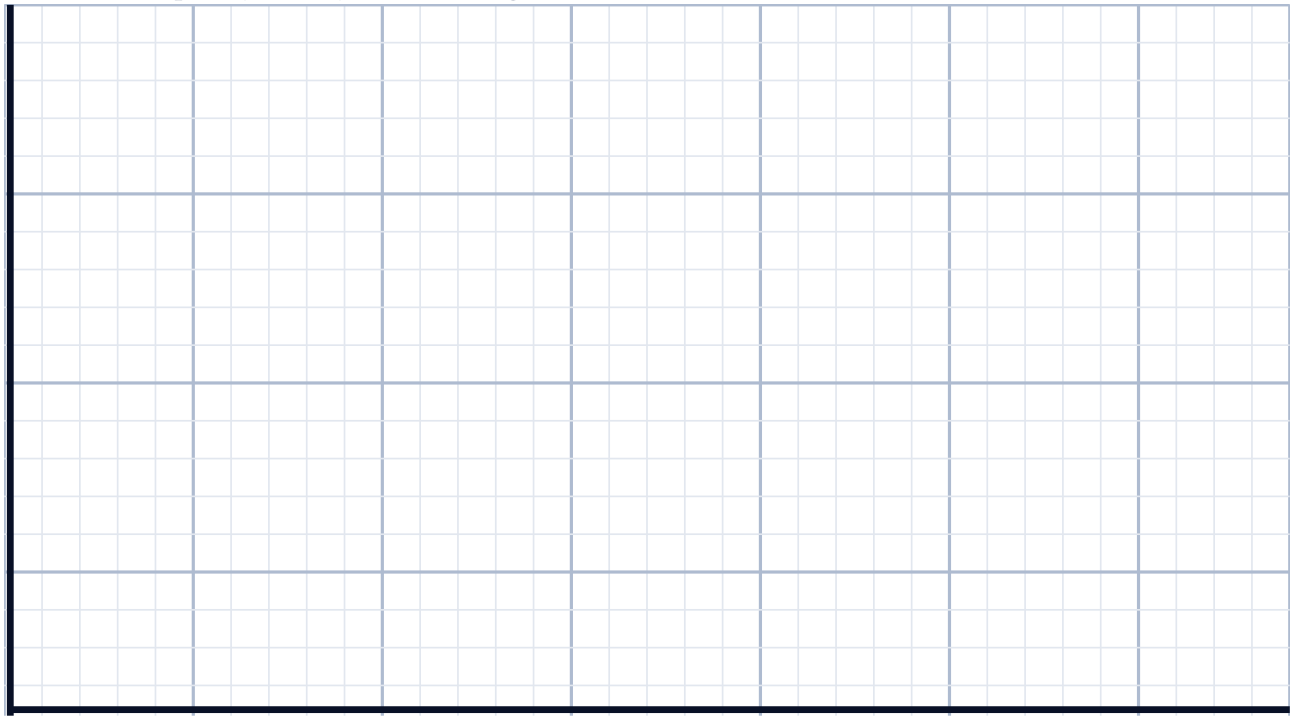
Body mass: plot the group pattern

Graph type: Histogram · continuous measurement; bars touch.

Title: Distribution of

Choose equal-width mass intervals that cover the readings; show the pooled frequency, not individual people.

Vertical axis: Frequency (anonymous readings)



Horizontal axis: Mass interval (kg)

Use equal steps; begin the count scale at zero.

Frequency table

Mass interval (kg)	Tally	n
_____		
_____		
_____		
_____		
_____		
_____		

Think about the evidence

1. Which mass interval has the greatest frequency?
2. What does the shape or spread show about this voluntary sample?
3. What can a class histogram not tell us about any one person?

Check: Are the axes named and scaled? Do the bars touch for continuous values and have gaps for categories? Did you plot only anonymous group counts?

INVESTIGATION 02 / METHOD + RECORD

Height

LEARNING OBJECTIVE

Measure height in centimetres using a consistent, respectful method, then describe the distribution of volunteer measurements with a histogram.



Image: Variation Lab Live website

Method

1. Taking part is optional. Do not comment on or compare anyone's body.
2. Secure a height measure on a level floor; keep the area clear. Never climb.
3. Stand comfortably upright. Follow footwear directions only if comfortable; the participant may position the headpiece. Ask first; never touch.
4. Read at eye level; record anonymous centimetres only (80–230 cm). Use the practice dataset if uncomfortable.

Risk assessment Hazard → safe action

Unwanted contact	Ask permission; let the participant position the headpiece; allow a pass.
Unstable equipment	Secure it on level flooring; keep the area clear; never climb.

Record your data Up to 40 anonymous readings · 80–230 cm. Row numbers are not people; keep values private.

No.	Height (cm)	No.	Height (cm)
1		21	
2		22	
3		23	
4		24	
5		25	
6		26	
7		27	
8		28	
9		29	
10		30	
11		31	
12		32	
13		33	
14		34	
15		35	
16		36	
17		37	
18		38	
19		39	
20		40	

INVESTIGATION 02 / GRAPH + INTERPRET

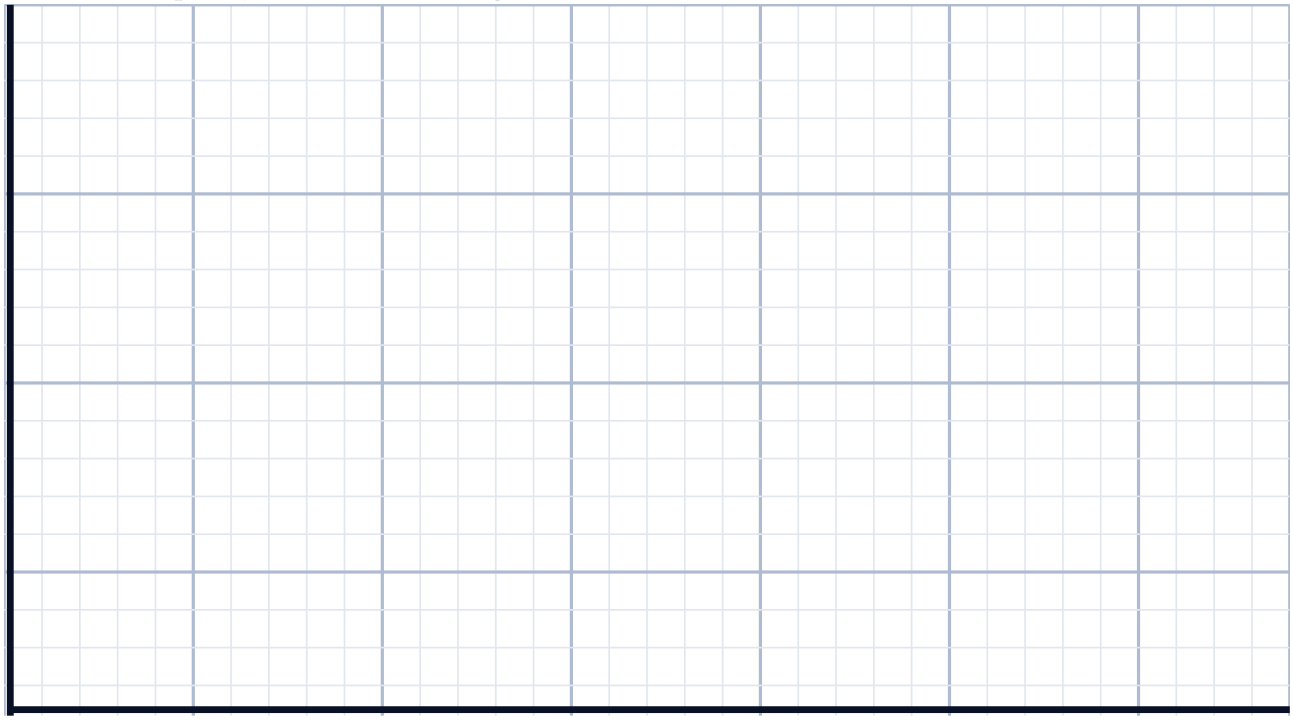
Height: plot the group pattern

Graph type: Histogram · continuous measurement; bars touch.

Title: Distribution of

Choose equal-width height intervals and use one boundary rule consistently so every reading is counted once.

Vertical axis: Frequency (number of readings)



Horizontal axis: Height interval (cm)

Use equal steps; begin the count scale at zero.

Frequency table

Height interval (cm)	Tally	n
_____		
_____		
_____		
_____		
_____		
_____		

Think about the evidence

1. Which interval contains the most readings?
2. What could make height measurements less consistent?
3. How might voluntary participation limit this sample?

Check: Are the axes named and scaled? Do the bars touch for continuous values and have gaps for categories? Did you plot only anonymous group counts?

INVESTIGATION 03 / METHOD + RECORD

Hand span

LEARNING OBJECTIVE

Measure hand span safely in centimetres and use anonymous class data to describe its distribution with a histogram.



Image: Variation Lab Live website

Method

1. Participation is optional; use your own hand only.
2. Rest it on a flat surface and spread your fingers comfortably.
3. Measure thumb tip to little-finger tip in centimetres; do not force joints.
4. Record anonymous 5–35 cm. Use 5 cm classes; include 35 in the final class.

Risk assessment Hazard → safe action

Finger discomfort	Stop if it hurts; spread only as far as comfortable.
Shared ruler	Clean it between users; keep it away from the mouth.

Record your data Up to 40 anonymous readings · 5–35 cm; use pooled totals only.

No.	Hand span (cm)	No.	Hand span (cm)
1		21	
2		22	
3		23	
4		24	
5		25	
6		26	
7		27	
8		28	
9		29	
10		30	
11		31	
12		32	
13		33	
14		34	
15		35	
16		36	
17		37	
18		38	
19		39	
20		40	

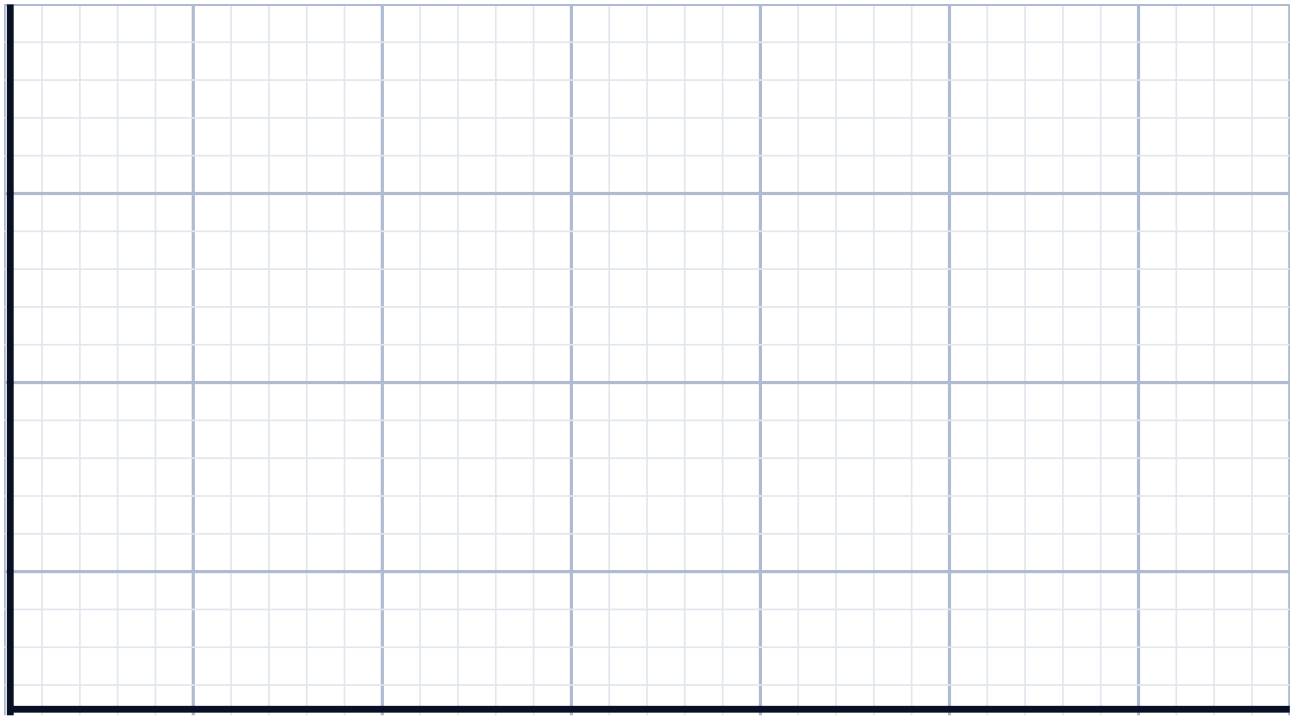
INVESTIGATION 03 / GRAPH + INTERPRET

Hand span: plot the group pattern

Graph type: Histogram · continuous measurement; **Title:** Distribution of bars touch.

Use the same-width 5 cm intervals shown in the frequency table; start the frequency scale at zero.

Vertical axis: Frequency (anonymous measurements)



Horizontal axis: Hand span interval (cm)

Use equal steps; begin the count scale at zero.

Frequency table

Hand span (cm)	Tally	n
5–<10		
10–<15		
15–<20		
20–<25		
25–<30		
30–35		

Think about the evidence

1. Which hand-span interval has the greatest frequency?
2. Does the pattern look even, clustered, or spread out?
3. What could affect accuracy or limit how representative this group is?

Check: Are the axes named and scaled? Do the bars touch for continuous values and have gaps for categories? Did you plot only anonymous group counts?

INVESTIGATION 04 / METHOD + RECORD

Tongue rolling

LEARNING OBJECTIVE

Record voluntary, anonymous self-reports of tongue-rolling ability and explain why a class pattern cannot establish genetic cause or destiny.



Image: Variation Lab Live website

Method

1. Choose Can, Cannot, or pass; no explanation is needed.
2. Self-report only; do not test, touch, or inspect anyone else.
3. Tally anonymous Can/Cannot totals; a pass is not counted.
4. Use separate bars. This is not a one-gene test and predicts no one's genes.

Risk assessment Hazard → safe action

Pressure or teasing	Allow a quiet pass; never rank or single anyone out.
Unwanted contact	No touching or shared objects; count anonymous totals only.

Record your data Anonymous totals only; a pass is not counted.

Category	Tally marks	Count
Can		
Cannot		

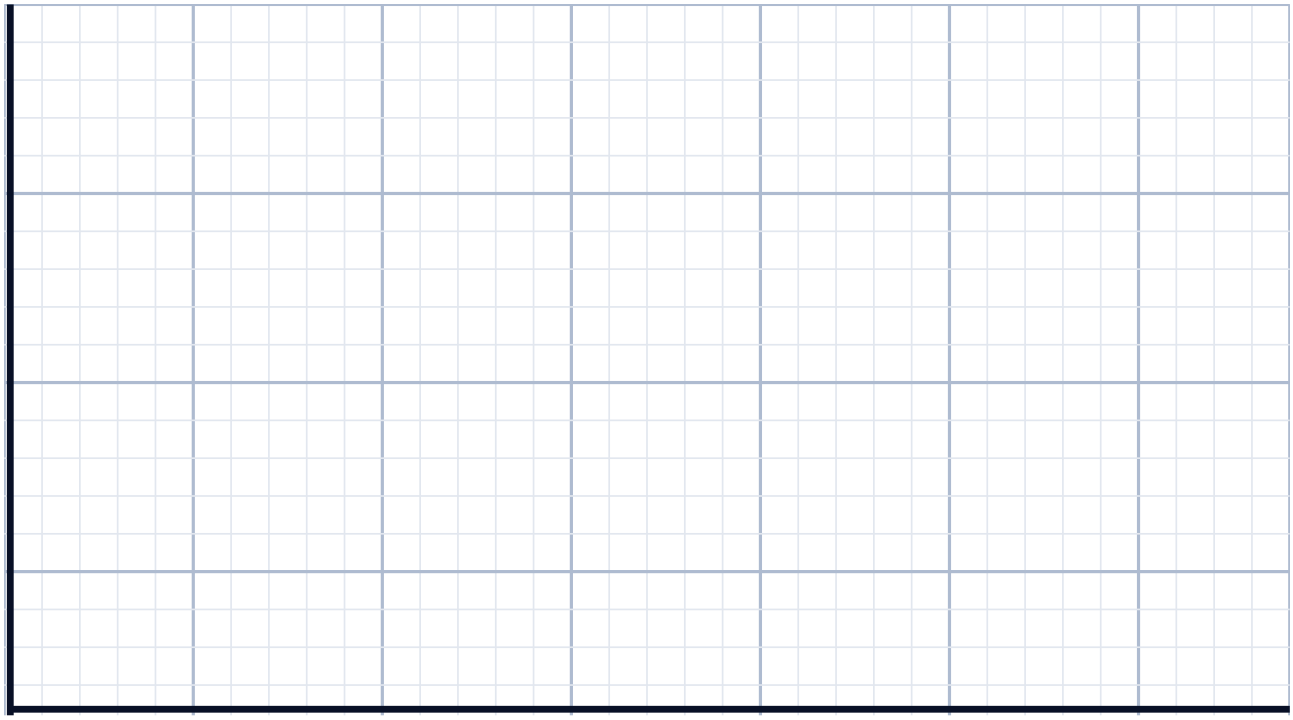
INVESTIGATION 04 / GRAPH + INTERPRET

Tongue rolling: plot the group pattern

Graph type: Separated bar chart · categorical data; **Title:** Distribution of leave gaps.

Draw one separate bar for Can and one for Cannot. A gap shows that the labels are distinct categories.

Vertical axis: Number of voluntary responses



Horizontal axis: Self-reported category

Use equal steps; begin the count scale at zero.

Frequency table

Category	Tally	n
Can		
Cannot		

Think about the evidence

1. Which category has the larger pooled count, or are they equal?
2. Why can these totals not identify anyone's genes?
3. What limits how far this class pattern can be generalised?

Check: Are the axes named and scaled? Do the bars touch for continuous values and have gaps for categories? Did you plot only anonymous group counts?

INVESTIGATION 05 / METHOD + RECORD

Eye colour

LEARNING OBJECTIVE

Classify voluntary eye-colour observations into the workbook's categories and graph anonymous totals without making genetic or health conclusions.



Image: Variation Lab Live website

Method

1. Choose self-report, brief self-observation, or pass.
2. Use room light only; no flash. Skip if unsure or uncomfortable.
3. Select Brown, Blue, Green, or Hazel; do not force a category.
4. Tally anonymous totals only. Do not take photos or make health claims.

Risk assessment Hazard → safe action

Bright light	Use room light only; no flash or directed beam.
Privacy	Self-report is fine; show pooled category totals only.

Record your data Use Brown, Blue, Green, or Hazel; no photos or identifiers.

Category	Tally marks	Count
Brown		
Blue		
Green		
Hazel		

INVESTIGATION 05 / GRAPH + INTERPRET

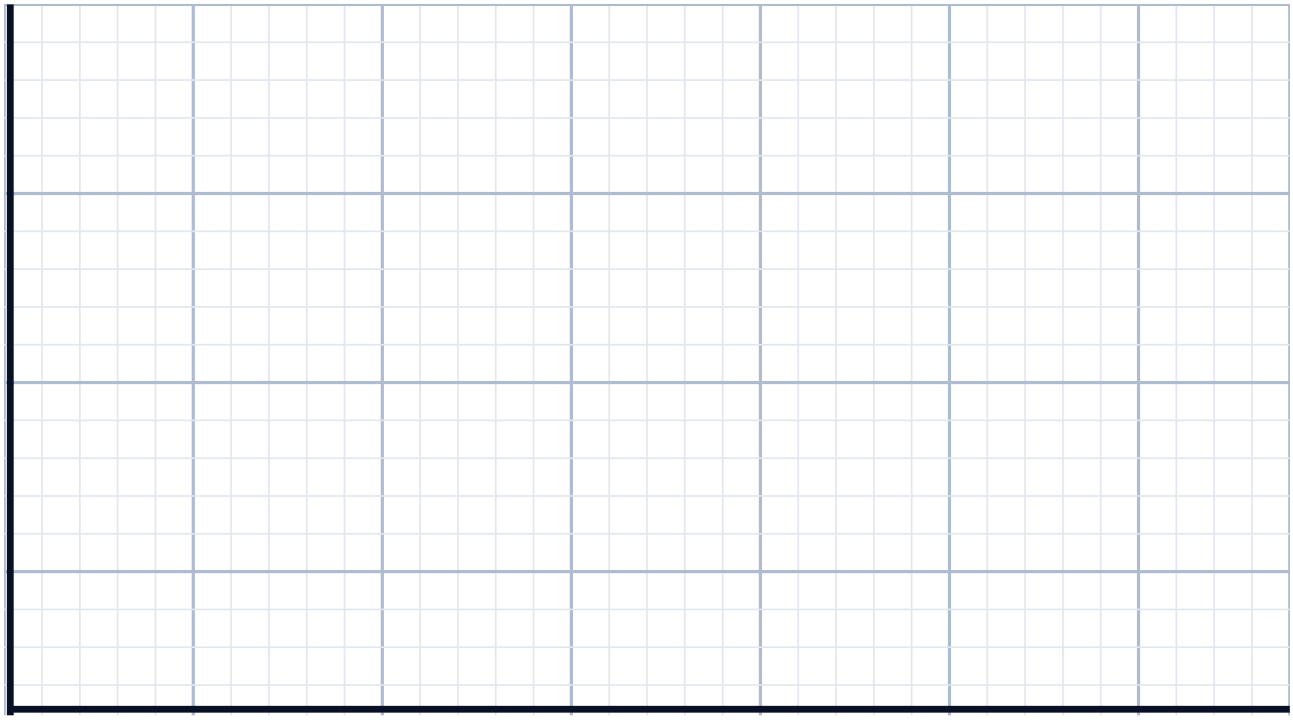
Eye colour: plot the group pattern

Graph type: Separated bar chart · categorical data; **Title:** Distribution of leave gaps.

Use four distinct bars, one per category, with gaps.

Start the count axis at zero and use equal steps.

Vertical axis: Number of anonymous responses



Horizontal axis: Eye-colour category

Use equal steps; begin the count scale at zero.

Frequency table

Category	Tally	n
Brown		
Blue		
Green		
Hazel		

Think about the evidence

1. Which eye-colour category has the largest pooled count?
2. Why should the chart not be used as a health test?
3. Why does a category tally not prove a simple genetic cause?

Check: Are the axes named and scaled? Do the bars touch for continuous values and have gaps for categories? Did you plot only anonymous group counts?

INVESTIGATION 06 / METHOD + RECORD

Simulated blood group

LEARNING OBJECTIVE

Use fictional anti-A and anti-B reaction patterns to classify simulated cards as A, B, AB, or O, then graph the category counts.



Image: Variation Lab Live website

Method

1. Use teacher-provided fictional cards only; no blood or sharps.
2. For each card, record anti-A and anti-B as clumps/no clumps.
3. Classify: A = anti-A only; B = anti-B only; AB = both; O = neither.
4. Tally four groups. Label the chart SIMULATED; it is not a medical test.

Risk assessment Hazard → safe action

Blood or sharps	Simulation only; no real blood, lancets, or bodily fluids.
Medical disclosure	Use fictional card codes; never request real results.

Record your data Fictional cards only. No people, names, diagnoses, or real results.
Simulation key: A = anti-A only · B = anti-B only · AB = both · O = neither. Write "clumps" or "no clumps" in the two reaction columns.

Fictional card	Anti-A	Anti-B	ABO class
F-1			
F-2			
F-3			
F-4			
F-5			
F-6			
F-7			
F-8			
F-9			
F-10			
F-11			
F-12			

Category	Tally marks	Count
A		
B		
AB		
O		

INVESTIGATION 06 / GRAPH + INTERPRET

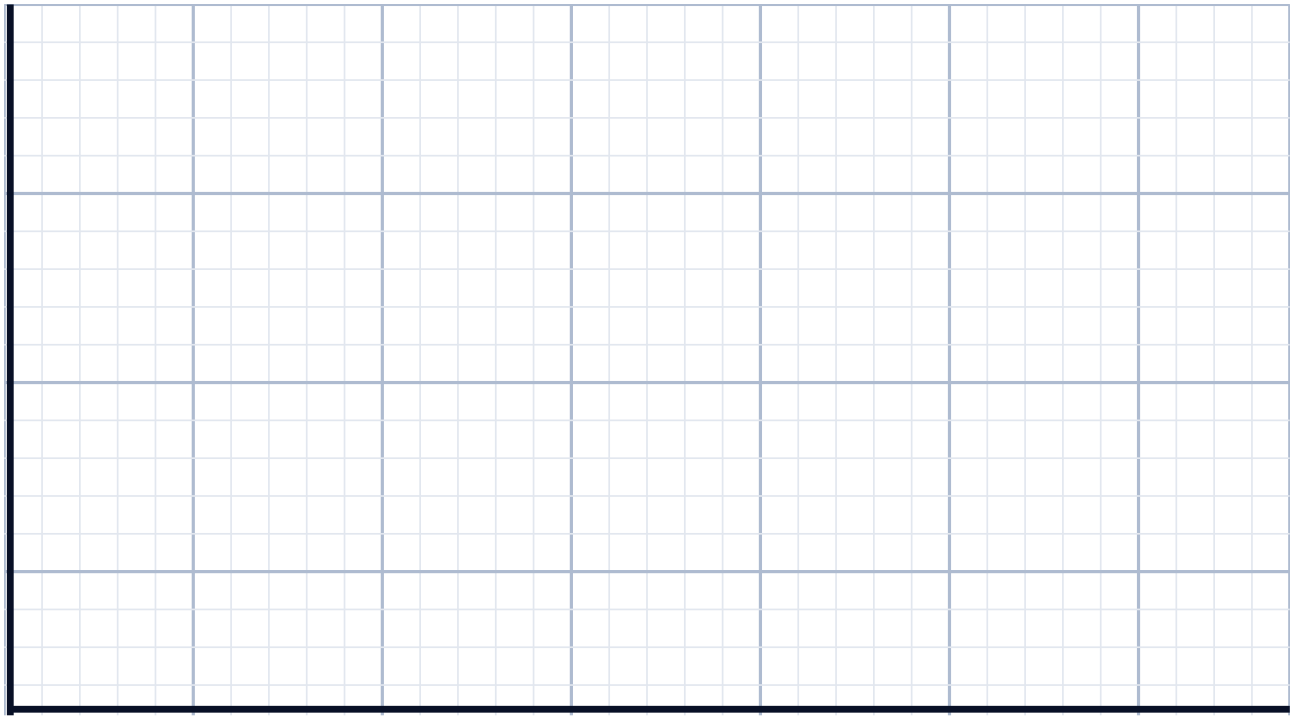
Simulated blood group: plot the group pattern

Graph type: Separated bar chart · simulated categories; leave gaps.

Title: Distribution of

Use one separate bar for A, B, AB, and O, including a zero-count category. Mark the title as SIMULATED.

Vertical axis: Number of fictional cards



Horizontal axis: Simulated ABO category

Use equal steps; begin the count scale at zero.

Frequency table

Simulated type	Tally	n
A		
B		
AB		
O		

Think about the evidence

1. Which simulated category has the greatest count?
2. Do the four frequencies add to the number of fictional cards?
3. Why can this classroom activity not diagnose or identify a real person?

Check: Are the axes named and scaled? Do the bars touch for continuous values and have gaps for categories? Did you plot only anonymous group counts?

INVESTIGATION 07 / METHOD + RECORD

Leaf length

LEARNING OBJECTIVE

Measure leaf lengths consistently and use a histogram to describe how a set of continuous measurements varies.



Image: Variation Lab Live website

Method

1. Use a teacher-approved, known non-toxic species; otherwise use practice data.
2. Measure along the blade's longest central line from base to tip; exclude the stalk.
3. Measure in place using one method and, if possible, one species. Do not pick plants.
4. Record up to 40 leaves (0.5–60 cm); avoid your face and wash hands.

Risk assessment Hazard → safe action

Plant irritants	Use a known approved plant; do not touch uncertain species.
Plant damage/ residue	Measure in place; avoid the face and wash hands.

Record your data Up to 40 leaves · 0.5–60 cm; same species/method; no identifiers.

No.	Length (cm)	No.	Length (cm)
1		21	
2		22	
3		23	
4		24	
5		25	
6		26	
7		27	
8		28	
9		29	
10		30	
11		31	
12		32	
13		33	
14		34	
15		35	
16		36	
17		37	
18		38	
19		39	
20		40	

INVESTIGATION 07 / GRAPH + INTERPRET

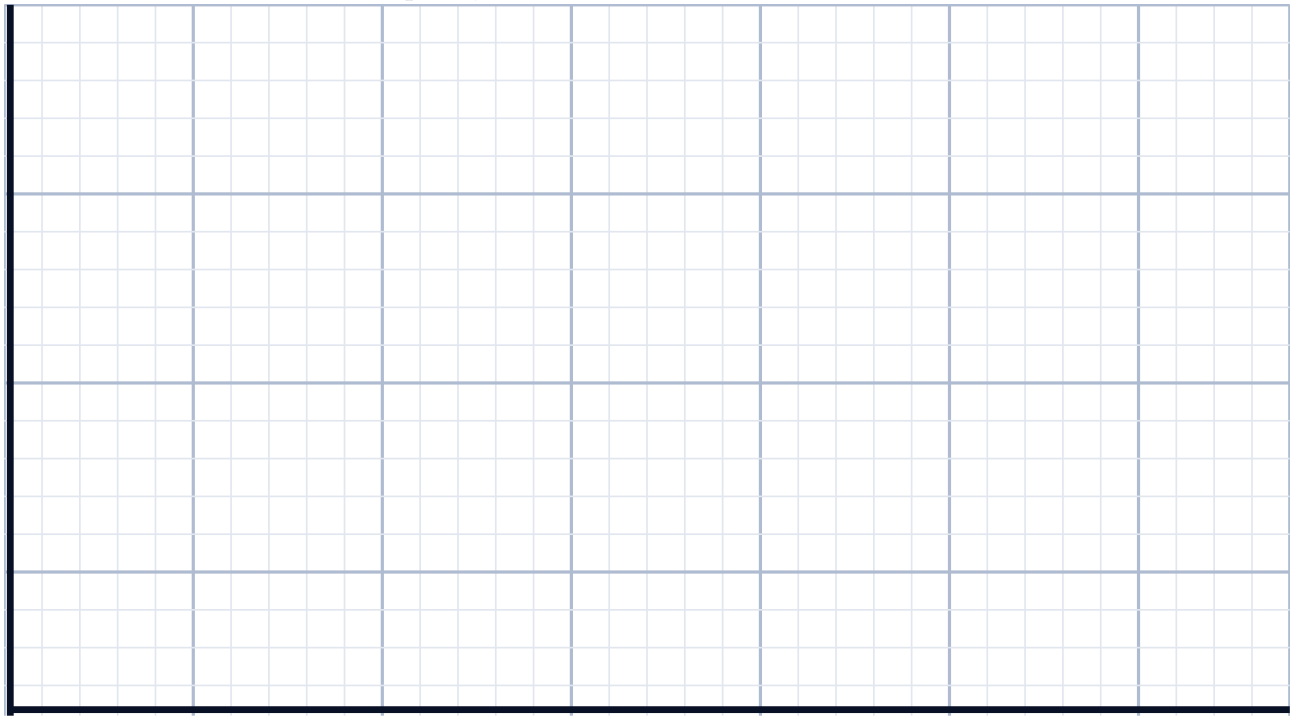
Leaf length: plot the group pattern

Graph type: Histogram · continuous measurement; bars touch.

Title: Distribution of

Choose equal-width length intervals that cover the leaves. Count each leaf once and make adjacent bars touch.

Vertical axis: Number of leaves (frequency)



Horizontal axis: Leaf-length interval (cm)

Use equal steps; begin the count scale at zero.

Frequency table

Leaf length (cm)	Tally	n
_____		
_____		
_____		
_____		
_____		
_____		

Think about the evidence

1. Which interval contains the most leaves?
2. What variation do you notice within the measured set?
3. How could plant species, growth conditions, or measurement method affect the pattern?

Check: Are the axes named and scaled? Do the bars touch for continuous values and have gaps for categories? Did you plot only anonymous group counts?

INVESTIGATION 08 / METHOD + RECORD

Seed mass

LEARNING OBJECTIVE

Measure individual seed mass in grams and use a frequency histogram to describe variation in the sample.



Image: Variation Lab Live website

Method

1. Use teacher-approved, untreated seeds under school allergy rules; otherwise use practice data.
2. Tare a clean, dry boat if directed; weigh one seed at a time in grams.
3. Record up to 40 values (0.001–10 g); check outliers, never alter them.
4. Do not eat seeds or make dust; clean up as directed.

Risk assessment Hazard → safe action

Allergy or treated seed	Check school rules; use known approved untreated seeds only.
Dust or choking	Do not eat; keep away from mouths; avoid dust and tidy.

Record your data Up to 40 seeds · 0.001–10 g; consistent precision; no student details.

No.	Mass (g)	No.	Mass (g)
1		21	
2		22	
3		23	
4		24	
5		25	
6		26	
7		27	
8		28	
9		29	
10		30	
11		31	
12		32	
13		33	
14		34	
15		35	
16		36	
17		37	
18		38	
19		39	
20		40	

INVESTIGATION 08 / GRAPH + INTERPRET

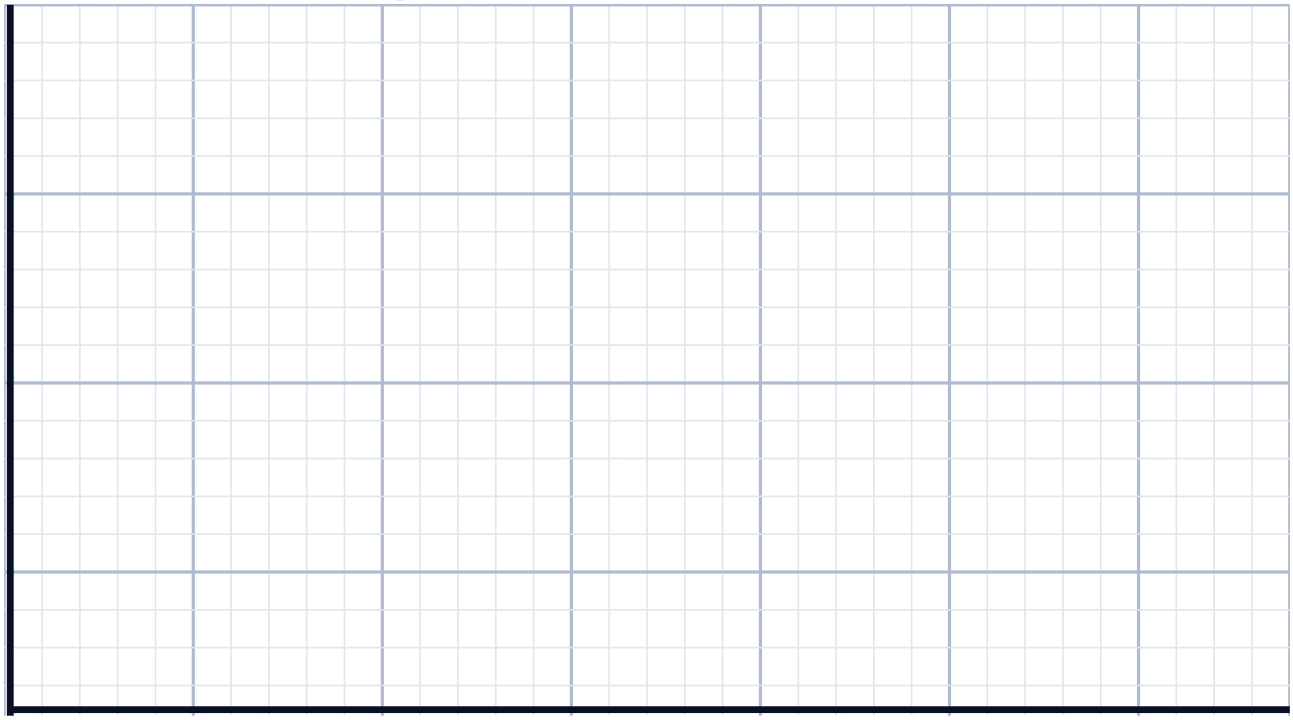
Seed mass: plot the group pattern

Graph type: Histogram · continuous measurement; bars touch.

Title: Distribution of

Choose equal-width mass intervals that cover the seed readings. Count each seed once; adjacent bars touch.

Vertical axis: Number of seeds (frequency)



Horizontal axis: Seed-mass interval (g)

Use equal steps; begin the count scale at zero.

Frequency table

Seed mass (g)	Tally	n
_____		
_____		
_____		
_____		
_____		
_____		

Think about the evidence

1. Which mass interval contains the most seeds?
2. What does the histogram show about the spread of this sample?
3. How could balance precision or seed choice affect the results?

Check: Are the axes named and scaled? Do the bars touch for continuous values and have gaps for categories? Did you plot only anonymous group counts?